Spatio-Temporal Probabilistic Forecasting of Photovoltaic Power Based on Monotone Broad Learning System and Copula Theory

Nan Zhou, Xiaoyuan Xu, *Member, IEEE*, Zheng Yan, *Member, IEEE*, and Mohammad Shahidehpour, *Life Fellow, IEEE*

***Abstract*—Probabilistic forecasting of photovoltaic (PV) power provides system operators with pertinent information on the uncertainty of PV power generation. This paper proposes a spatio-temporal probabilistic forecasting model based on monotone broad learning system (MBLS) and Copula theory. MBLS is a novel neural network structure for providing an efficient quantile regression solution. MBLS guarantees the monotonicity between quantiles and their probability for thoroughly avoiding the quantile crossing problem. The historical PV data are then clustered using the self-organizing map and samples in each cluster are used for Copula parameter estimations. The proposed approach provides an efficient spatio-temporal forecast of multiple PV plants by combining marginal distributions predicted by MBLS with Copula functions. The real-world data of PV plants in Australia and USA are used to the validate the superiority of the proposed method through detailed comparisons with existing methods using comprehensive evaluation criteria. The presented results demonstrate that the proposed method can provide high-quality probabilistic forecasts corresponding with PV power scenarios.**

***Index Terms*—Photovoltaic power forecasting, Monotone broad learning system, Copula theory, quantile regression.**

I. Introduction

RENEWABLE energy generation such as photovoltaic (PV) power has been developing rapidly in recent years. While effectively alleviating environmental concerns, the high penetration of PV power into power systems challenges secure and stable operations of power systems [1]. The volatile and intermittent nature of PV power has led to potentially adverse effects on voltage rise, protection arrangements, and power quality [2]. In this context, PV power forecasting is essential for reliable and secure power system operations.

Manuscript received 31 October 2021; revised 18 February 2022 and 5 April 2022; accepted 6 May 2022. Date of publication 10 May 2022; date of current version 20 September 2022. This work was supported by the National Natural Science Foundation of China under Grants U2166201 and 52077136. Paper no. TSTE-01100-2021. (*Corresponding author: Xiaoyuan Xu.*)

Nan Zhou, Xiaoyuan Xu, and Zheng Yan are with the Key Laboratory of Control of Power Transmission and Conversion, Shanghai Jiao Tong University, Ministry of Education, Shanghai 200240, China (e-mail: zhounan2015@sjtu.edu.cn; xuxiaoyuan@sjtu.edu.cn; yanz@sjtu.edu.cn).

Mohammad Shahidehpour is with the Galvin Center for Electricity Innovation, Illinois Institute of Technology, Chicago, IL 60616 USA, and also with the Center of Research Excellence in Renewable Energy and Power Systems, King Abdulaziz University, Jeddah 21589, Saudi Arabia (e-mail: ms@iit.edu).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TSTE.2022.3174012.

Digital Object Identifier 10.1109/TSTE.2022.3174012

Probabilistic forecasting techniques, which quantify the uncertainty of prediction targets, have been widely used in renewable energy power forecasting. Broadly speaking, probabilistic forecasting techniques are classified into parametric and nonparametric methods [3]. Although the former only needs a small amount of computation, improper assumptions on probability distributions may yield huge errors in forecasting results [4]. Nonparametric methods, on the other hand, do not make any assumptions on the type of probability distribution functions. Popular nonparametric probabilistic forecasting methods include kernel density estimation (KDE) [5], [6], Bayesian learning [7], quantile regression forest (QRF) [8], and quantile regression (QR) [4], [9]–[14].

In recent years, QR-based probabilistic forecasting is intensively studied since it can be flexibly incorporated into various machine learning methods. Reference [4] has proposed a probabilistic forecasting method of PV power based on QR and extreme learning machine (ELM) and used the repository-based particle swarm optimization to train the prediction model. In [9], [10], a direct quantile regression (DQR) model based on QR and ELM is formulated as a linear programming (LP) problem with high computational efficiency. In [11], different QR-based methods including quantile regression neural network (QRNN) and quantile regression gradient boosting (QRGB) are integrated to improve the probabilistic forecasting performance. Recently, the combination of QR and deep learning is discussed by many investigators. References [12] and [13] have proposed the improved QRNN (iQRNN) using popular techniques in deep learning such as batch training and skip connections. The pinball loss function of QR is applied in [14] to train the long short-term memory (LSTM) neural network, which produces reliable probabilistic forecasts of load demands.

Although QR has been successfully applied in probabilistic forecasting of power systems, two issues have not been fully addressed in the existing literature. First, deep learning tends to suffer from the time-consuming training process and parameter tuning difficulties despite its powerful capability to handle high-volume data. Second, the quantile crossing phenomenon may occur in forecasting results, which means that the predicted quantiles do not increase with the specified probability [15]. Common practices to prevent quantile crossing include: 1) naive rearrangement of quantiles according to their probability [11], 2) adding penalty terms of crossed quantiles in the loss function

1949-3029 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

[4]; 3) simultaneously estimating multiple quantiles with additional constraints ensuring non-crossing [9], [10]. Since naive rearrangement is a compromise to achieve monotonicity, the methods 2 and 3 are adopted by many investigators. However, as pointed out by [11], quantile crossing still happens in the prediction stage because these two methods only consider the non-crossing constraints in the training stage.

The above forecasting models focus on individual renewable energy generation. Note that geographically close PV units exhibit strong dependence in power output since they are influenced by similar weather conditions [16]. To evaluate such spatio-temporal dependence, the vector autoregression (VAR) [17], multivariate generalized autoregressive conditional heteroskedasticity (mGARCH) model [18] and Copula function [5], [19]–[21] have been applied to probabilistic forecasting or scenario generation of renewable power. The autoregressive methods such as VAR and mGARCH require parametric marginal distributions, while Copula functions also work for nonparametric marginal distributions.

In [5], [19], the marginal probability distributions of wind power are obtained from KDE and QR, respectively, and then the spatial dependence of multiple wind farms is modeled based on pair-Copula functions. In [20], QR is combined with bidirectional LSTM to perform probabilistic forecasting, and the joint probability distribution is formulated using empirical Copula. In [21], marginals of wind power are modeled by the Gaussian mixture model and the spatio-temporal model is constructed based on the Copula theory. Although Copula has been widely used to model renewable power dependence, the conventional practice is to estimate parameters of a single Copula function based on all the historical data, which neglects conditional correlations in different external conditions.

Recently, the broad learning system (BLS) was proposed by Chen, *et al.* [22]. Different from deep neural networks that improve the approximation capability by increasing the number of layers, BLS expands the width of a single hidden layer to enhance its generalization performance [23]. Besides, A. J. Cannon proposed monotone composite quantile regression neural network (MCQRNN) [24]. The structure of MCQRNN ensures that predicted quantiles are monotonically increasing with their probability. Inspired by the idea of MCQRNN, we propose a spatio-temporal probabilistic forecasting model of PV power which applies a neural network called monotone broad learning system (MBLS) to tackle the quantile crossing problem. MBLS is trained efficiently and shows a comparable approximation capability with deep neural networks. In addition, to establish the spatio-temporal forecasting models in different weather conditions, the self-organizing map (SOM) is used to cluster historical data and different Copula functions are estimated based on cluster samples. Finally, marginal distributions obtained by MBLS are combined with the spatio-temporal dependence models established by Copula to predict the joint probability distributions of the power outputs for different PV units and time steps. The main contributions of this paper are as follows:

1) We propose MBLS to predict the marginal probability distributions of PV power. We give the theoretical proof on the monotonicity of prediction targets with respect to prediction inputs; thus, MBLS fundamentally overcomes the quantile crossing problem of existing QR-based probabilistic forecasting methods.
2) We design prediction models considering the PV power characteristics. The ramp function is applied to the output layer of MBLS to guarantee that the predicted quantiles lie within the nominal range of PV power. The Huber norms are employed to approximate the non-differentiable functions in the QR loss function, and the training of MBLS is deduced based on back propagation. The designed network improves the accuracy and computational efficiency of the forecasting model.
3) We combine SOM with Copula functions to establish the joint probability distributions of PV power. SOM is used to cluster the PV power and meteorological data, and the Copula function is established for each cluster to model the spatio-temporal dependence of multiple PV units and time steps for various external conditions.
4) The validity of the proposed method is verified using real-world PV data by comparing with existing methods. The QRMBLS method provides reliable and sharp probabilistic forecasts without quantile crossing in an efficient manner, and its overall performance is better than benchmark methods. Moreover, the proposed spatio-temporal forecasting method generates scenarios that reflect spatio-temporal dependence in accordance with actual PV power data.

The rest of the paper is organized as follows. The methodologies of marginal probabilistic forecasting using MBLS and spatio-temporal forecasting using clustering-based Copula are presented in Sections II and III, respectively. Section IV introduces evaluation criteria of PV power forecasting. In Section V, comprehensive case studies and results are shown. Finally, conclusions are drawn in Section VI.

II. Probabilistic Forecasting Based on MBLS

This section introduces the QR-based probabilistic forecasting model and proposes the MBLS network to tackle the quantile crossing problem of QR methods. The proposed probabilistic forecasting model is called QRMBLS for brevity.

A. Quantile Regression

The prediction target of PV power at time t is denoted as $y(t)$, and the input vector of the prediction model is denoted as $\boldsymbol{x}(t)$. Suppose that $\boldsymbol{x}(t)$ is a v-dimensional vector, i.e., $\boldsymbol{x}(t) = [x_1(t), x_2(t), \ldots, x_v(t)]$. Then, the dataset for PV power forecasting is presented as

$$D = \{x(t), y(t)\}_{t=1}^{T} \tag{1}$$

where T is the number of samples.

To estimate the probability distribution of $y(t)$, the quantile $q^{\tau}(t)$ with probability $\tau \in [0, 1]$ is defined as

$$q^{\tau}(t) = F_t^{-1}(\tau) \tag{2}$$

where F_t is the cumulative distribution function (CDF) of $y(t)$.

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

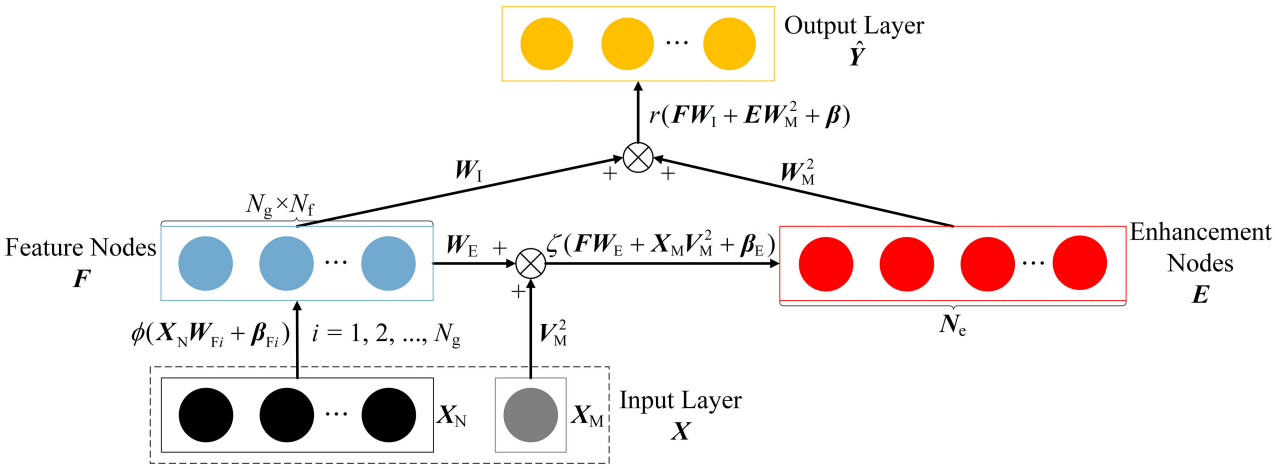


Fig. 1. Network structure of MBLS.

The QR methods approximate the conditional probability distribution of random variables in the form of quantiles [25]. Specifically, conditional quantiles are estimated by minimizing the sum of asymmetrically weighted absolute deviations:

$$L_\tau = \frac{1}{T}\sum_{t=1}^{T}\rho_\tau(y(t) - \hat{q}^\tau(t)) \tag{3}$$

where $\hat{q}^\tau(t)$ is the predicted conditional quantile, and $\rho_\tau(\cdot)$ is the following pinball loss function:

$$\rho_\tau(\varepsilon) = \begin{cases} \tau\varepsilon, & \varepsilon \geq 0 \\ (\tau-1)\varepsilon, & \varepsilon < 0 \end{cases} \tag{4}$$

Based on the predicted quantiles, the prediction interval (PI) with the nominal coverage probability $1-\gamma$ is obtained as

$$\mathrm{PI}^{1-\gamma}(t) = [\hat{q}^{\underline{\tau}}(t), \hat{q}^{\bar{\tau}}(t)] \tag{5}$$

where the upper and lower bounds of PI are the quantiles with probabilities $\bar{\tau}$ and $\underline{\tau}$. The central PI satisfies

$$\underline{\tau} = 1 - \bar{\tau} = \gamma/2 \tag{6}$$

B. Network Structure of MBLS

MBLS is developed to predict PV power marginal probability distributions with monotonically increasing quantiles. Different from deep neural networks, MBLS expands the width of a single hidden layer to enhance its generalization performance. The network structure of MBLS is illustrated in Fig. 1 and presented as follows.

First, the input matrix $\boldsymbol{X} \in \mathbb{R}^{N\times u}$ is divided into two parts: $\boldsymbol{X}_\mathrm{M} \in \mathbb{R}^{N\times m}$ contains input variables with monotonically increasing relationship with the output $\hat{Y}$, and $\boldsymbol{X}_\mathrm{N} \in \mathbb{R}^{N\times n}$ contains input variables without monotonicity constraints. Specifically, to avoid quantile crossing in probabilistic forecasting, it is desired that the quantiles are monotonically increasing with their probabilities. Hence, $\boldsymbol{X}_\mathrm{N}$ is the input samples of the prediction model, which consists of $\boldsymbol{x}(t)$ in (1); $\boldsymbol{X}_\mathrm{M}$ is the probabilities of corresponding quantiles. The input matrices are given by (7):

$$\boldsymbol{X}_\mathrm{M} = \begin{bmatrix} \tau_1 \\ \cdots \\ \tau_1 \\ \tau_2 \\ \cdots \\ \tau_2 \\ \cdots \\ \tau_K \\ \cdots \\ \tau_K \end{bmatrix}, \quad \boldsymbol{X}_\mathrm{N} = \begin{bmatrix} x_1(1) & \ldots & x_v(1) \\ \cdots & \cdots & \cdots \\ x_1(T) & \ldots & x_v(T) \\ x_1(1) & \ldots & x_v(1) \\ \cdots & \cdots & \cdots \\ x_1(T) & \ldots & x_v(T) \\ \cdots & \cdots & \cdots \\ x_1(1) & \ldots & x_v(T) \\ \cdots & \cdots & \cdots \\ x_1(T) & \ldots & x_v(T) \end{bmatrix} \tag{7}$$

Second, for N_g feature mapping groups and N_f feature nodes in each group, the feature nodes in the i_th feature mapping group are generated by extracting the feature of $\boldsymbol{X}_\mathrm{N}$, as stated below:

$$\boldsymbol{F}_i = \phi(\boldsymbol{X}_\mathrm{N}\boldsymbol{W}_{\mathrm{F}i} + \boldsymbol{\beta}_{\mathrm{F}i}), \quad i = 1, 2, \ldots, N_\mathrm{g} \tag{8}$$

where the weighting matrix $\boldsymbol{W}_{\mathrm{F}i}$ and bias term $\beta_{\mathrm{F}i}$ of feature nodes are randomly generated; the mapping function ϕ is established using the sparse autoencoder [22].

Third, N_e enhancement nodes are constructed based on feature mappings $\boldsymbol{F} = [\boldsymbol{F}_1, \boldsymbol{F}_2, \ldots, \mathrm{F}_{Ng}]$ and $\boldsymbol{X}_M$:

$$\boldsymbol{E} = \zeta(\boldsymbol{F}\boldsymbol{W}_\mathrm{E} + \boldsymbol{X}_\mathrm{M}\boldsymbol{V}_\mathrm{M}^2 + \boldsymbol{\beta}_\mathrm{E}) \tag{9}$$

where the weighting matrix $\boldsymbol{W}_\mathrm{E}$ and bias term $\boldsymbol{\beta}_\mathrm{E}$ of enhancement nodes are randomly generated. The activation function $\zeta(\cdot)$ is the hyperbolic tangent function in this paper.

Finally, the hidden layer of MBLS contains both the feature and enhancement nodes, which are connected to the output layer. Therefore, the output of MBLS is given by

$$\hat{\boldsymbol{Y}} = r(\boldsymbol{F}\boldsymbol{W}_\mathrm{N} + \boldsymbol{E}\boldsymbol{W}_\mathrm{M}^2 + \boldsymbol{\beta}) \tag{10}$$

where the bias term $\boldsymbol{\beta} = [\beta, \beta, \ldots, \beta]^\mathrm{T}$ is an N-dimensional vector; $r(\cdot)$ is a ramp function restricting quantiles to interval $[0, P_\mathrm{cap}]$:

$$r(\pi) = \begin{cases} 0, & \pi < 0 \\ \pi, & 0 \leq \pi \leq P_\mathrm{cap} \\ P_\mathrm{cap}, & \pi > P_\mathrm{cap} \end{cases} \tag{11}$$

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

where P_{cap} denotes the rated capacity of PV units.

For PV power probabilistic forecasting, the outputs $\hat{Y}$ are the predicted quantiles. When predicting the quantiles with probabilities $\tau_1 < \tau_2 < \ldots < \tau_K$, the prediction target $\boldsymbol{Y}$ and MBLS output $\hat{\boldsymbol{Y}}$ are designed as

$$\boldsymbol{Y} = [y(1), \ldots (T), y(1), \ldots (T), \ldots, y(1), \ldots (T)]^{\mathrm{T}} \quad (12)$$

$$\hat{\boldsymbol{Y}} = [\hat{q}^{\tau_1}(1), \ldots, \hat{q}^{\tau_1}(T), \hat{q}^{\tau_2}(1), \ldots, \hat{q}^{\tau_2}(T), \ldots, \hat{q}^{\tau_K}(1), \ldots, \hat{q}^{\tau_K}(T)]^{\mathrm{T}} \quad (13)$$

For simplicity, if $\boldsymbol{M}$ is a matrix, then $\boldsymbol{M}_i$ denotes the i_{th} row of $\boldsymbol{M}$ while $M_{i,j}$ denotes the element at the i_{th} row and j_{th} column of $\boldsymbol{M}$. If $\boldsymbol{v}$ is a column vector, then v_i denotes the i_{th} element of $\boldsymbol{v}$. We obtain Lemma 1 based on the structure of MBLS.

Lemma 1: For any $n_1, n_2 \in \{1, 2, \ldots, N\}$, if $\boldsymbol{X}_{\mathrm{N}\,n1} = \boldsymbol{X}_{\mathrm{N}\,n2}$ and $X_{\mathrm{M}\,n1} \geqslant X_{\mathrm{M}\,n2}$, then $\hat{Y}_{n_1} \geq \hat{Y}_{n_2}$.

The proof of Lemma 1 is given in Appendix A. According to (7), the elements of $\boldsymbol{X}_{\mathrm{N}}$ remain the same in predicting the quantiles of PV power at one time step. Hence, a higher quantile probability leads to a larger predicted quantile based on Lemma 1; thus, the monotonicity of quantiles is satisfied.

C. Monotone Broad Learning System Training

To predict K quantiles of T time steps, we modify the loss function of (3) as

$$L = \frac{1}{N}\sum_{n=1}^{N} \rho_{\tau_n}(Y_n - \hat{Y}_n), \tau_n = X_{\mathrm{M}n} \quad (14)$$

Since the pinball loss function (4) and the ramp function (11) are not differentiable everywhere, we establish the Huber-norm versions of (4) and (10) to make the loss function differentiable [14]. The approximated pinball loss and ramp functions are denoted by ${\rho_\tau}^{(\mathrm{A})}(\varepsilon)$ and $r^{(\mathrm{A})}(\pi)$, of which the detailed formulas are given in Appendix B. In addition, regularization terms are considered in the loss function to prevent overfitting. Finally, the loss function is stated as

$$L^{(\mathrm{A})} = \frac{1}{N}\sum_{n=1}^{N} \rho_{\tau_n}^{(\mathrm{A})}(Y_n - \hat{Y}_n^{(\mathrm{A})}) + \lambda_{V_{\mathrm{M}}}\frac{1}{N_{\mathrm{e}}}\sum_{i=1}^{N_{\mathrm{e}}} V_{\mathrm{M}1,i}^2 + \lambda_{W_{\mathrm{M}}}\frac{1}{N_{\mathrm{e}}}\sum_{i=1}^{N_{\mathrm{e}}} W_{\mathrm{M}i}^2 + \lambda_{W_{\mathrm{N}}}\frac{1}{N_{\mathrm{g}}N_{\mathrm{f}}}\sum_{i=1}^{N_{\mathrm{e}}} W_{\mathrm{N}i}^2 \quad (15)$$

where $\lambda_{V_{\mathrm{M}}}, \lambda_{W_{\mathrm{M}}}$ and $\lambda_{W_{\mathrm{N}}}$ are the regularization parameters; $\hat{Y}_n^{(\mathrm{A})}$ is the n_{th} element of the approximated output $\hat{\boldsymbol{Y}}^{(\mathrm{A})}$:

$$\hat{\boldsymbol{Y}}^{(\mathrm{A})} = r^{(\mathrm{A})}(\boldsymbol{F}\boldsymbol{W}_{\mathrm{N}} + \boldsymbol{E}\boldsymbol{W}_{\mathrm{M}}^2 + \boldsymbol{\beta}) \quad (16)$$

The parameters of MBLS, including $\boldsymbol{V}_{\mathrm{M}}$, $\boldsymbol{W}_{\mathrm{M}}$, $\boldsymbol{W}_{\mathrm{N}}$, $\boldsymbol{\beta}_{\mathrm{E}}$, and $\boldsymbol{\beta}$, are estimated using the gradient-based Adam algorithm [26]. The gradient of $L^{(\mathrm{A})}$ is calculated via backpropagation, which is derived in Appendix C.

III. Spatio-Temporal Forecasting Based on Copula

This section first utilizes SOM to obtain different PV power data clusters and then models the spatio-temporal dependence of PV power in each cluster based on the Copula theory. Finally, the framework of spatio-temporal PV power forecasting model is summarized.

Algorithm 1: Training of Self-Organizing Map.

Input: input samples $\boldsymbol{\chi}(t)$ ($t = 1, 2, \ldots, T$); maximum number of iterations $\bar{K}$; learning rate $\alpha(k)$; neighborhood radius $\sigma(k)$.
Output: weight vectors of SOM $\boldsymbol{w}_i$ ($i = 1, 2, \ldots, N_{\mathrm{O}}$)
1: $k \leftarrow 1$
2: Initialize $\boldsymbol{w}_i$ using random values $w_i^{(1)}$
3: **while** $k \leq \bar{K}$ **do**
4: Select a sample $\boldsymbol{\chi}^{(k)}$ randomly from the input
5: Normalize $\boldsymbol{w}_i$ and $\boldsymbol{\chi}^{(k)}$
6: $\boldsymbol{w}_{\mathrm{W}}^{(k)} \leftarrow \arg\min_{\boldsymbol{w}_i^{(k)}} ||\boldsymbol{\chi}^{(k)} - \boldsymbol{w}_i^{(k)}||$
7: $\boldsymbol{w}_i^{(k+1)} \leftarrow \boldsymbol{w}_i^{(k)} + \alpha(k)\exp\left(-\frac{||\boldsymbol{w}_{\mathrm{W}}^{(k)} - \boldsymbol{w}_i^{(k)}||}{2\sigma^2(k)}\right) \times (\boldsymbol{\chi}^{(k)} - \boldsymbol{w}_i^{(k)})$
8: $k \leftarrow k + 1$
9: **end while**
10: $\boldsymbol{w}_i \leftarrow \boldsymbol{w}_i^{(\bar{K})}$

A. Self-Organizing Map

SOM is a neural network-based clustering technique that transforms the original input space into the bi-dimensional output space [27]. The input sample $\boldsymbol{\chi}(t) \in \mathbb{R}^{NI}$ of SOM is the combination of the input vectors of the prediction models of different PV units. The output layer consists of $N_{\mathrm{O}} = N_{\mathrm{R}} \times N_{\mathrm{C}}$ neurons arranged in N_{R} rows and N_{C} columns. These neurons compete to become activated, and there is only one active neuron, called 'winner neuron', for each input sample $\boldsymbol{\chi}(t)$. The weight vector of each neuron in the output layer is denoted as $\boldsymbol{w}_i \in \mathbb{R}^{NI}$ ($i = 1, 2, \ldots, N_{\mathrm{O}}$). The neuron with the weight vector closest to the input sample is chosen as the winner neuron $\boldsymbol{w}_{\mathrm{W}}$, which represents the input sample's cluster.

$$\boldsymbol{w}_{\mathrm{W}} = \arg\min_{\boldsymbol{w}_i} \|\boldsymbol{\chi}(t) - \boldsymbol{w}_i\| \quad (17)$$

The training procedure of SOM is given in Algorithm 1. SOM clusters the PV power dataset into several subsets with different external conditions, and then the spatio-temporal PV power dependence is obtained based on the historical PV power data of each subset.

B. Copula Theory

For N_{PV} PV units and N_{lead} step-ahead predictions, there are $N_{\mathrm{PV}} \times N_{\mathrm{lead}}$ random variables in the probability model of PV power forecasting. In this paper, we use the Copula theory to establish the joint probability distribution of multiple random variables and obtain the spatio-temporal dependence of PV power. Specifically, based on the Sklar's theorem, the joint probability distribution of random variables is established as [5],

$$F_{\mathrm{J}}(\xi_1, \xi_2, \ldots, \xi_d) = C(F_1(\xi_1), F_2(\xi_2), \ldots, F_d(\xi_d)) \quad (18)$$

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

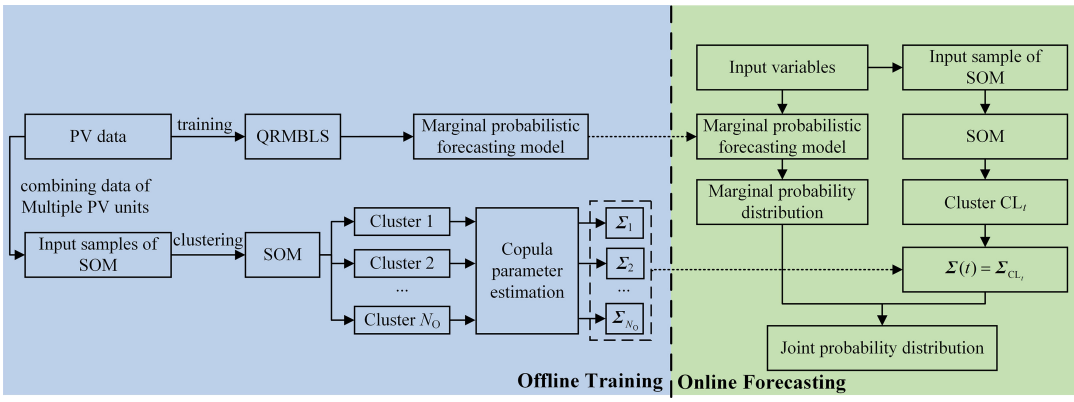


Fig. 2. Overall framework of the proposed probabilistic forecasting model.

where F_J is the joint CDF of multiple random variables; C is the Copula function; F_j ($j = 1, 2, \ldots, d$ and $d = N_{PV} \times N_{lead}$) is the marginal CDF of each random variable. In this paper, we use the Gaussian Copula to model the high-dimensional random variables as:

$$c(u_1, u_2, \ldots, u_d) = \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}\boldsymbol{z}^T(\boldsymbol{\Sigma} - \boldsymbol{I})\boldsymbol{z}\right) \quad (19)$$

where $u_j = F_j(\xi_j)$ and follows the uniform distribution on the interval [0, 1]; $\boldsymbol{\Sigma}$ is the correlation matrix of Gaussian Copula; $\boldsymbol{z} = (z_1, z_2, \ldots, z_d)^T$ and $z_j = \Phi^{-1}(u_j)$ with $\Phi(\cdot)$ as the CDF of the standard Normal distribution; $|\boldsymbol{\Sigma}|$ is the determinant of $\boldsymbol{\Sigma}$; $\boldsymbol{I}$ is the identity matrix. The marginal CDF $F_j(\xi_j)$ of each random variable is obtained based on the predicted quantiles and $\boldsymbol{\Sigma}$ is estimated with the maximum likelihood estimation algorithm [28] for each PV power data cluster.

C. Framework for Spatio-Temporal Forecasting of PV Power

The framework of the proposed PV power forecasting method is illustrated in Fig. 2. The procedure for offline model training is given as follows:

1) Training QRMBLS-based probabilistic forecasting models: The marginal probabilistic forecasting models of different PV units and look-ahead times are trained.
2) Training SOM and estimating Gaussian Copula parameters: To distinguish the spatio-temporal dependence of PV power in different meteorological conditions, the prediction inputs of individual PV units are combined to form input samples $\boldsymbol{\chi}(t)$. Then, SOM would divide the input samples into several clusters. Finally, based on the PV power data of each cluster, the parameter $\boldsymbol{\Sigma}$ of Gaussian Copula is estimated to represent the spatial-temporal dependence in different conditions.

The online prediction procedure is given as follows:

1) Predicting marginal probability distributions of PV power: For N_{PV} PV units and N_{lead} step-ahead predictions, the $N_{PV} \times N_{lead}$ marginal probability distributions of PV power are obtained via the pre-trained QRMBLS model.
2) Predicting spatio-temporal dependence of PV power: First, the input variables of PV power predictions are combined to form the input sample of SOM. Second, the cluster of the input sample is determined based on the pre-trained SOM. Third, the correlation matrix of Gaussian Copula is obtained for the selected cluster. Finally, the joint probability distribution of PV power is obtained by combining the marginal distributions with Gaussian Copula.

Based on the predicted probability distribution model, the spatio-temporal PV power samples are generated by sampling methods [29] to evaluate the accuracy of PV power forecasting.

IV. Evaluation Criteria of PV Power Forecasting

A. Evaluation Criteria of Marginal Probabilistic Forecasting

The performance of the marginal probabilistic forecasting method is evaluated in terms of reliability and sharpness [3].

1) Reliability: Reliability is measured based on the similarity between probability distributions of actual and predicted PV power. Here, we use the absolute deviation index to evaluate the reliability of probabilistic forecasting [4]:

$$\mathrm{Dev_k} = |\hat{\tau}_k - \tau_k| \quad (20)$$

where $\hat{\tau}_k$ is the empirical quantile probability:

$$\hat{\tau}_k = \frac{1}{T_{test}} \sum_{t=1}^{T_{test}} \eta^{\tau_k}(t) \quad (21)$$

and $\eta^{\tau_k}(t)$ is an indicator given by

$$\eta^{\tau_k}(t) = \begin{cases} 1, & y(t) \le \hat{q}^{\tau_k}(t) \\ 0, & y(t) > \hat{q}^{\tau_k}(t) \end{cases} \quad (22)$$

1) Sharpness: We use the average width of prediction intervals to evaluate sharpness as:

$$\bar{\delta}^{\gamma} = \frac{1}{T_{test}} \sum_{t=1}^{T_{test}} \left(\hat{q}^{1-\gamma/2}(t) - \hat{q}^{\gamma/2}(t)\right) \quad (23)$$

2) Comprehensive criteria: We use the skill score (SS) [4] and Winkler score (WS) [7] to comprehensively assess the reliability and the sharpness of forecasting results. SS

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

is the pinball loss calculated on the testing dataset:

$$\mathrm{SS} = \frac{1}{T_{\text{test}}K}\sum_{t=1}^{T_{\text{test}}}\sum_{k=1}^{K}(\tau_k - \eta^{\tau_k}(t))(y(t) - \hat{q}^{\tau_k}(t)) \tag{24}$$

WS is defined as

$$\mathrm{WS} = \frac{1}{T_{\text{test}}N_{\text{PI}}}\sum_{t=1}^{T_{\text{test}}}\sum_{i=1}^{N_{\text{PI}}}\mathrm{WS}_{\gamma_i}(t) \tag{25}$$

where N_{PI} is the number of PIs, $\mathrm{WS}_\gamma(t)$ is the Winkler score for PI with coverage rate $(1-\gamma)$ at time t, which is stated as

$$\mathrm{WS}_\gamma(t) = \begin{cases} \delta_\gamma(t), & \mathrm{LB}_\gamma(t) \le y(t) \le \mathrm{UB}_\gamma(t) \\ \delta_\gamma(t) + 2[\mathrm{LB}_\gamma(t) - y(t)]/\gamma, & y(t) < \mathrm{LB}_\gamma(t) \\ \delta_\gamma(t) + 2[y(t) - \mathrm{UB}_\gamma(t)]/\gamma, & y(t) > \mathrm{UB}_\gamma(t) \end{cases} \tag{26}$$

where $\mathrm{LB}_\gamma(t)$ and $\mathrm{UB}_\gamma(t)$ are the lower and upper bounds of PI, and $\delta_\gamma(t) = \mathrm{UB}_\gamma(t) - \mathrm{LB}_\gamma(t)$. Because both SS and WS are negatively oriented, a smaller score implies better performances.

1) Crossing loss: We use the crossing loss (CL) [30] to evaluate the quantile crossing issue of probabilistic forecasting results:

$$\mathrm{CL} = \frac{1}{T_{\text{test}}(K-1)}\sum_{t=1}^{T_{\text{test}}}\sum_{k=1}^{K-1}\max(0, \hat{q}^{\tau_k}(t) - \hat{q}^{\tau_{k+1}}(t)) \tag{27}$$

B. Evaluation Criteria of Spatio-Temporal Forecasting

We use the log score (LS) [17] and energy score (ES) [19] to evaluate the spatio-temporal probabilistic forecasting models. LS is the mean negative log likelihood of predicted distributions evaluated at measurements:

$$\mathrm{LS} = \frac{1}{T_{\text{test}}}\sum_{t=1}^{T_{\text{test}}} -\ln(\hat{f}_{\mathrm{J}\mathbf{t}}(\xi_1(t), \xi_2(t), \ldots, \xi_d(t))) \tag{28}$$

where $\hat{f}_{\mathrm{J}\mathbf{t}}(\cdot)$ is the predicted joint PDF at time t; $\xi_j(t)$ ($j = 1, 2, \ldots, d$) are the measurements of random variables at time t.

ES is calculated with measurements and predicted scenarios:

$$\mathrm{ES}(\boldsymbol{\Xi}(t), \boldsymbol{\xi}(t)) = \frac{1}{s}\sum_{i=1}^{s}\|\boldsymbol{\Xi}(t)_i - \boldsymbol{\xi}(t)\|_2 - \frac{1}{2s^2}\sum_{i=1}^{m}\sum_{j=1}^{m}\left\|\boldsymbol{\Xi}(t)_i - \boldsymbol{\Xi}(t)_j\right\|_2 \tag{29}$$

where $\boldsymbol{\xi}(t) \in \mathbb{R}^d$ represents the measurement vector; $\boldsymbol{\Xi}(t) \in \mathbb{R}^{d\times s}$ represents s scenarios, and $\boldsymbol{\Xi}(t)_i$ is the i_{th} column of $\boldsymbol{\Xi}(t)$. The elements of $\boldsymbol{\xi}(t)$ and $\boldsymbol{\Xi}(t)$ are normalized with respect to the capacity of PV units. The average ES in different time steps is

$$\mathrm{ES} = \frac{1}{T_{\text{test}}}\sum_{t=1}^{T_{\text{test}}}\mathrm{ES}(\boldsymbol{\Xi}(t), \boldsymbol{\xi}(t)) \tag{30}$$

Note that both LS and ES are negatively oriented.

TABLE I
BASIC INFORMATION ON PV FORECASTING DATASETS

Season	*Case* 1		*Case* 2	
	Summer	Winter	Spring	Fall
Beginning of Daytime	5:50	7:10	5:35	6:30
End of Daytime	19:50	18:40	18:45	17:50
Number of Days	166	175	273	273
Sample Number in Training Dataset	28080	23250	31110	25071
Sample Number in Testing Dataset	3600	3000	15300	12330
N_{L}	24	18	18	17

V. CASE STUDIES

A. Dataset Description

The dataset of five PV units in Yulara, Australia [31] (denoted as *Case* 1) is used to test the effectiveness of the proposed forecasting method. The five PV units are denoted as PV1, PV2, PV3, PV4, and PV5, with rated capacities 226.8 kW, 22.6 kW, 38.3 kW, 327.6 kW, and 105.9 kW, respectively. The meteorological data includes solar irradiance, temperature, and wind speed measured at the Desert Gardens Weather Station in Yulara. The data resolutions of both meteorological data and PV power are 5 minutes. This paper focuses on the intra-hour forecasting [3]. Hence, we predict the PV power from five minutes to one hour ahead. In this case, $N_{\text{PV}} = 5$ and $N_{\text{lead}} = 12$, and there are 60 marginal distributions in the spatio-temporal forecasting.

Table I presents information on PV power forecasting data. Because the pattern of PV power varies in different seasons, the data in summer (January, February, and December) and winter (June, July, and August) of the years 2017 and 2018 are used in case studies, resulting in two separate datasets corresponding to the two seasons. For data cleansing, the dates with more than 10 abnormal data are removed. For other dates with invalid data, linear interpolation is used to rectify the dataset. Hence, for each season, the data in the last 20 days are treated as the testing data and the rest are treated as the training data. Only the data during daytime are preserved since the power output of PV units is zero at night. In addition, the sample size and day numbers after data cleansing are given in Table I. The input variables of prediction models include PV power observations up to N_{L} lag time steps, lag temperature, wind speed and solar irradiance [4], [32]. In this paper, N_{L} is determined using the five-fold cross-validation with the BLS-based point forecasting model.

To strengthen the validity of the proposed QRMBLS method, another dataset of a distributed PV unit in Ashland, USA [33] (denoted as *Case* 2) is used in the case study of marginal probabilistic forecasting. The rated capacity of this PV is 15 kW, and the meteorological data includes solar irradiance, temperature, and wind speed. The data in spring (March, April, and May) and fall (September, October, November) of the years 2015, 2016, and 2017 are used. For each season, the data in the last 90 days are treated as the testing data and the rest are treated as the training data. The data resolution is also 5 minutes, and the dataset is preprocessed similarly to that in *Case* 1. The basic information of this dataset is also presented in Table I. The source

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

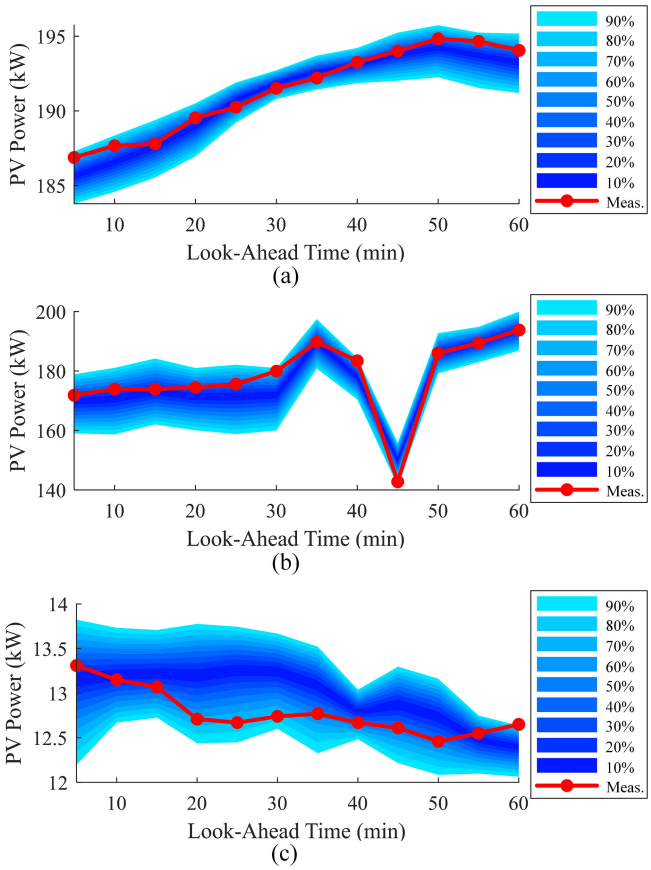


Fig. 3. Intra-hour prediction intervals obtained by QRMBLS at 12:00 p.m. on multiple dates: (a) Dec. 23, 2018, *Case* 1, (b) Dec. 12, 2018, *Case* 1. (3) Mar. 2, 2017, *Case* 2.

code for marginal probabilistic forecasting and spatio-temporal forecasting of *Case* 1 is available at [34].

B. Results of Marginal Probabilistic Forecasting

In the case studies of marginal probabilistic forecasting of *Case* 1, we only present the PV1 results since the other PV units are similar. To verify the superiority of the proposed QRMBLS model, existing methods including the persistence method [3], [10], QRNN [35], DQR [10], iQRNN [12] and MCQRNN [24] are used for comparison. The persistence method is a popular benchmark for ultra-short-term renewable energy power generation forecasting, where the underlying probability distribution is assumed to be the Gaussian distribution. The other methods, on the contrary, are non-parametric models. Specifically, several QRNN models are trained to predict the quantiles with different probabilities, which neglect the quantile crossing problem. The non-crossing quantile constraints are considered in the DQR model training. The MCQRNN network structure is designed to avoid the quantile crossing problem.

Fig. 3 shows the intra-hour prediction intervals of PV power at 12:00 p.m. under different conditions. The PIs cover most of the measured PV power outputs, which indicates the reliability of QRMBLS models. Moreover, these PIs are not symmetric

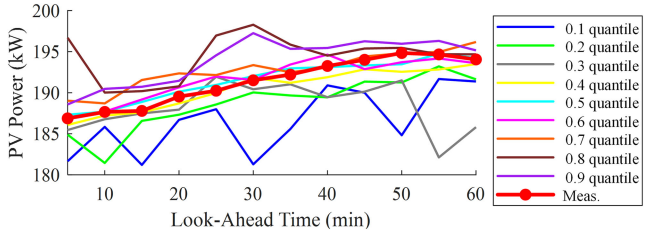


Fig. 4. Intra-hour predicted quantiles obtained by QRNN at 12:00 p.m. on Dec. 23, 2018, *Case* 1.

TABLE II
PERFORMANCE OF DIFFERENT METHODS IN THE SUMMER CASE

Method	5-minute ahead				
	Dev	$\bar{\delta}$ (kW)	SS (kW)	WS (kW)	CL (kW)
QRMBLS	**0.0405**	7.2940	1.7640	25.7535	**0**
Persistence	0.0449	9.6607	2.1017	28.6121	**0**
QRNN	0.0520	44.3146	4.0056	62.2279	11.5746
DQR	0.0963	**2.5624**	**1.7391**	**24.6504**	0.0038
iQRNN	0.0824	13.5069	2.4435	36.2744	0.0370
MCQRNN	0.0616	5.5982	2.1794	29.6804	**0**
Method	1-hour ahead				
	Dev	$\bar{\delta}$ (kW)	SS (kW)	WS (kW)	CL (kW)
QRMBLS	**0.0420**	7.6632	**2.7674**	39.3493	**0**
Persistence	0.2101	9.6607	11.8709	157.8993	**0**
QRNN	0.0475	47.0195	5.6025	79.2645	6.8671
DQR	0.1592	**4.1600**	2.8992	**37.2855**	0.0025
iQRNN	0.1396	13.6597	3.3310	47.5741	0.1263
MCQRNN	0.0705	7.6577	3.1938	43.5360	**0**

around the actual measurements; thus, parametric distributions such as a Gaussian distribution cannot accurately model the PV power probability distribution. In addition, PI widths on cloudy days are larger than those on sunny days because the PV power presents more volatility on cloudy days.

According to Fig. 3, the quantiles obtained by QRMBLS are monotonically increasing. However, the QRNN results suffer from the quantile crossing problem, as shown in Fig. 4. The crossed quantiles do not provide correct PIs. For example, in the 1-hour ahead forecasting results, the PI with a 40% coverage is wider than that of 80%, which contradicts the PI definition. Therefore, it is important to tackle the quantile crossing issue in probabilistic forecasting.

Table II gives the evaluation criteria of the 5-minute and 1-hour ahead probabilistic forecasting results for several methods. The proposed QRMBLS method shows the best performance in terms of reliability and gives similar results as MCQRNN in terms of sharpness. As to comprehensive criteria, QRMBLS ranks first and second in terms of the skill score in the 1-hour and 5-minute cases, respectively; it also ranks second in terms of the Winkler score. The persistence method performs well in the 5-minute case because it is the benchmark for very short lead time [4]. However, its performance deteriorates considerably in the 1-hour case representing the worst reliability and comprehensive scores. The QRNN model shows a good performance in terms of reliability, but it obtains the worst sharpness score among all the methods. In contrast, the DQR method exhibits good performances in sharpness and Winker scores but shows

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPUTATION TIME OF MODEL TRAINING

Method	*Case* 1		*Case* 2	
	Summer	Winter	Spring	Fall
QRMBLS	25.71 s	33.94 s	79.58 s	81.01 s
Persistence	0.61 s	0.88 s	0.30 s	0.92 s
QRNN	645.26 s	423.11 s	836.64 s	544.11 s
DQR	874.89 s	882.35 s	1838.38 s	2799.82 s
iQRNN	1144.35 s	926.66 s	1826.48 s	1757.93 s
MCQRNN	784.82 s	324.89 s	953.06 s	659.75 s

the worst reliability. The iQRNN and MCQRNN methods are inferior to the QRMBLS method in terms of reliability and comprehensive scores.

The CL scores are zero for the persistence method, MCQRNN and QRMBLS, which means that these methods guarantee the quantile monotonicity. This is because the persistence method uses a Gaussian distribution for modeling the predictive distribution, and MCQRNN and QRMBLS overcome the quantile crossing problem based on their network structures. QRNN and iQRNN obtain crossed quantiles because the quantiles of probabilities are predicted independently. The CL score of DQR is also nonzero because it only considers the non-crossing quantile constraints in the training stage.

The computational efficiency of QRMBLS is compared with those of benchmark methods in Table III. All the methods are implemented with MATLAB R2021a on a PC with Intel Core i5-8400 CPU @ 2.88 GHz and 8 GB RAM. The model training of the persistence method is extremely fast because it only estimates the parameters of Gaussian distributions. QRMBLS ranks second in terms of computation efficiency and it is about 25 times faster than QRNN in the summer case. QRNN estimates different quantiles using multiple neural networks, thus suffering from the heavy computational burden. DQR is also time-consuming because it solves a large-scale linear programming problem considering both the quantile range and the non-crossing quantile constraints. iQRNN costs a long time in model training because it is based on the deep neural networks of two hidden layers. The training time of MCQRNN is much longer than that of QRMBLS due to its much larger parameter numbers.

Figs. 5 and 6 present the radar charts of evaluation criteria to intuitively compare the performances of different methods. Considering that smaller evaluation criteria indicate better forecasting results, smaller areas of radar charts imply better performances. As shown in the figures, the proposed QRMBLS method obtains the best overall results for different cases, seasons and time horizons.

In conclusion, QRMBLS provides reliable probabilistic forecasts without quantile crossing, and its sharpness and comprehensive scores are comparable to those of state-of-the-art methods. The QRMBLS model is trained efficiently and presents high potentials for practical intra-hour PV power forecasting.

C. Results of Spatio-Temporal Forecasting

The QRMBLS method is employed in this section to obtain the marginal distributions for various PV plants and time steps.

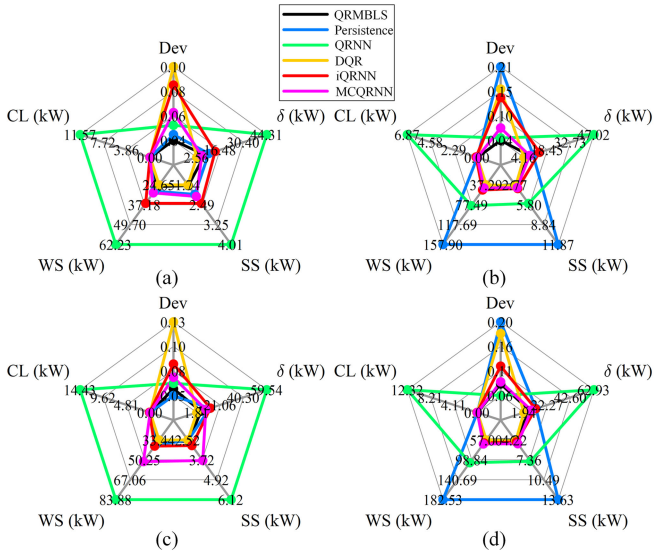


Fig. 5. Radar charts of evaluation criteria of different probabilistic forecasting methods in *Case* 1: (a) summer season, 5-min ahead, (b) summer season, 1-h ahead, (c) winter season, 5-min ahead, (d) winter season, 1-h ahead.

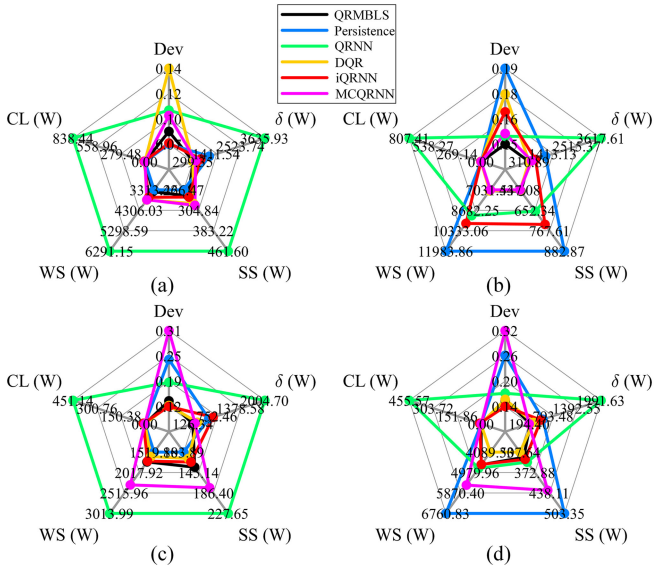


Fig. 6. Radar charts of evaluation criteria of different probabilistic forecasting methods in *Case* 2: (a) spring season, 5-min ahead, (b) spring season, 1-h ahead, (c) fall season, 5-min ahead, (d) fall season, 1-h ahead.

SOM is used to cluster the historical data for modeling the PV power dependences in prevailing conditions. The number of neurons in the SOM output layer, which is also the number of clusters, is 5×3 for summer and 4×3 for winter. Fig. 7 presents the samples of clusters 1 and 2 in summer. The results show that samples in the same cluster have similar PV power patterns, thus exhibiting similar spatio-temporal correlations.

The spatio-temporal forecasting model is compared with three methods to verify its superiority. The first one is the independent method, which neglects the PV power dependence to obtain

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

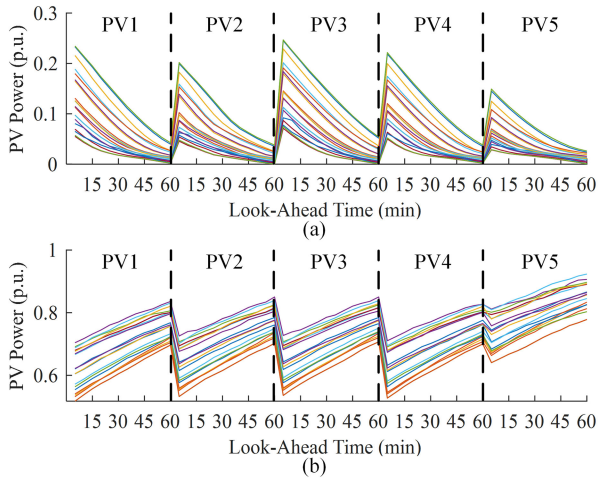


Fig. 7. SOM clustering results in summer: (a) samples in cluster 1, (b) samples in cluster 2.

TABLE IV
PERFORMANCES OF DIFFERENT COPULA-BASED METHODS

Method		Log Score		Energy Score (kW)	
		Summer	Winter	Summer	Winter
SOM-Copula	Gaussian Copula	137.5209	117.7809	**25.3443**	44.0178
	t Copula	**117.2484**	**106.7080**	25.6954	44.0398
Single-Copula	Gaussian Copula	143.9245	250.0262	25.6614	43.9167
	t Copula	132.7081	181.5174	25.6882	**43.6177**

scenarios from predictive marginal distributions. The second one is the single-Copula method, which models the spatio-temporal dependence with a single Copula function and estimates Copula parameters using all the data. The third one is the traditional VAR method, where the marginal distributions of PV power are assumed to be Gaussian.

Since both Gaussian Copula and t Copula can be used to model multivariate joint probability distributions, the performance of the two Copula functions is compared. However, the parameter estimation procedure of t Copula is quite time-consuming. Hence, only forecasting results of time intervals of 15 minutes, instead of 5 minutes, (the number of marginal distributions is 20 instead of 60) are presented.

The experimental results in Table IV show that the log scores of t Copula are smaller than those of Gaussian Copula, because t Copula can model the tail dependence of extreme power outputs of different PV units. However, the scenarios generated by the two Copula functions show similar energy scores, which implies that Gaussian Copula is comparable to t Copula in terms of spatio-temporal forecasting performances. Table V gives the time to estimate parameters of different Copula functions with different numbers of marginal distributions. Gaussian Copula can be estimated efficiently even for a large number of marginal distributions. In contrast, as the number of marginal distributions increases, the computation time of t Copula-based methods increases drastically. Hence, only Gaussian Copula is applied in subsequent case studies considering the computational efficiency issue.

TABLE V
COMPUTATION TIME OF COPULA PARAMETER ESTIMATION IN SUMMER CASE

Number of marginals		5	10	15	20
SOM-Copula	Gaussian Copula	0.0125 s	0.0203 s	0.0259 s	0.0777 s
	t Copula	64.94 s	2574 s	18335 s	59323 s
Single-Copula	Gaussian Copula	0.0043 s	0.0074 s	0.0124 s	0.0135 s
	t Copula	24.59 s	1522 s	9220 s	61047 s

TABLE VI
PERFORMANCES OF SPATIO-TEMPORAL FORECASTING METHODS

Method	Log Score		Energy Score (kW)	
	Summer	Winter	Summer	Winter
SOM-Copula	**34.5564**	**16.8648**	**38.6898**	**57.2706**
Single-Copula	49.1932	26.1366	44.5333	57.9717
Independent method	175.6649	175.7351	40.2602	59.7832
VAR	327.1987	332.5297	47.6855	58.7694

Table VI presents the experimental results of spatio-temporal forecasting when the number of marginal distributions is 60. The proposed SOM-Copula method obtains the best results in terms of both the log score and the energy score. The single-Copula method ranks second among the four methods. The independent method has very large scores because it assumes that the power outputs of PV units are independent, which is different from the actual PV power data. The VAR method has the largest log score because it assumes marginals to be Gaussian, which is not the case as discussed in the case study of marginal probabilistic forecasting.

Fig. 8 illustrates the 20 forecasting scenarios generated from the forecasting models. Although the marginal distributions of the SOM-Copula, single-Copula and independent methods are the same, the generated scenarios are significantly different. The forecasting scenarios of SOM-Copula are quite close to the actual PV power curves. The results of single-Copula present similar patterns as the actual curves, but they do not trace PV power fluctuations, such as the power output of PV2. This is because single-Copula does not distinguish different spatio-temporal dependences; thus, it cannot reflect the special variability pattern of PV power. The forecasting scenarios of the independent and VAR methods exhibit frequent and irregular fluctuations even when the actual PV power curves change steadily. Their worse performances are caused by the neglection of spatio-temporal dependence and wrong assumptions on marginal distributions, respectively.

To further validate the quality of forecasting scenarios, the correlation matrices of historical PV power data and forecasting scenarios generated by SOM-Copula are presented in Fig. 9. The correlation matrix is divided into 5×5 submatrices to reflect the spatial correlations of 5 PV units, and the elements of each submatrix describe the temporal correlation of the power output of each PV unit. The elements near the principal diagonals of submatrices are close to 1 because the PV power of consecutive time steps are strongly correlated. Meanwhile, the correlation

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

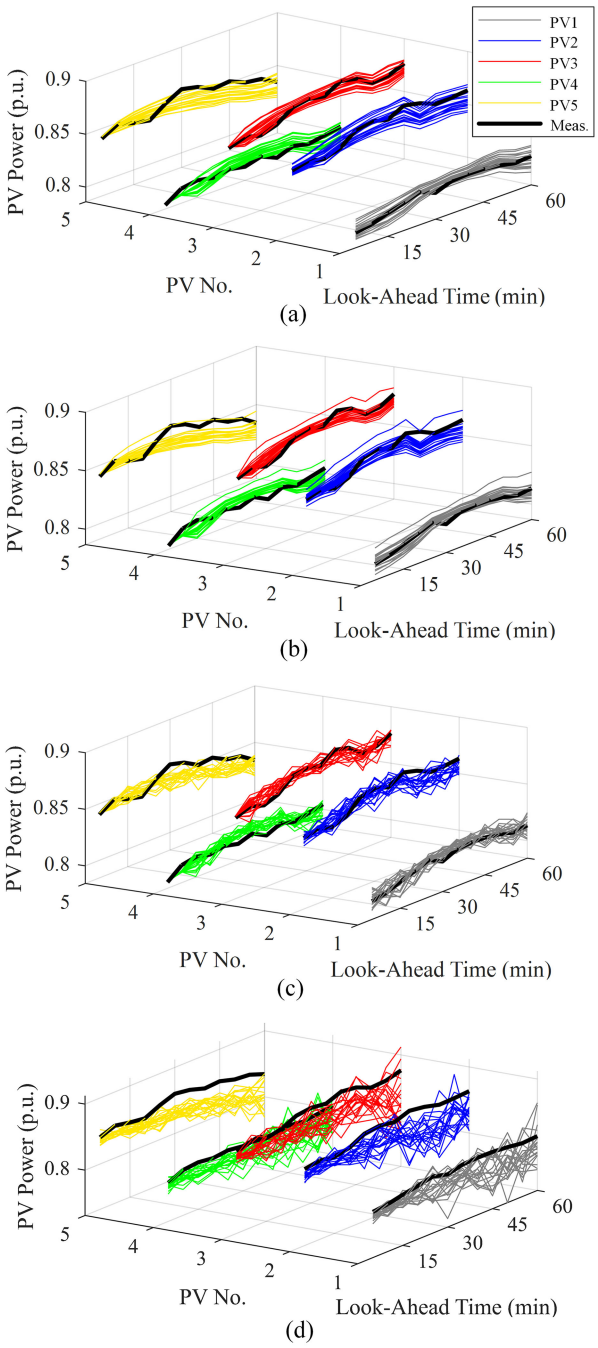


Fig. 8. Forecasting scenarios generated by different methods for 12:00 p.m., Dec. 23, 2018: (a) SOM-Copula method, (b) single-Copula method, (c) independent method, (d) VAR method.

coefficients decrease to around 0.8 for elements at the ends of secondary diagonals, which means that the temporal correlation becomes smaller for longer timespans. Fig. 9 shows that the spatio-temporal dependence of forecasting scenarios agrees with that of PV power measurements.

In summary, the proposed SOM-Copula method obtains accurate spatio-temporal dependences for multiple PV units and time steps. Once such correlation is modeled, high-quality forecasting scenarios can be generated to guide power system optimal operation.

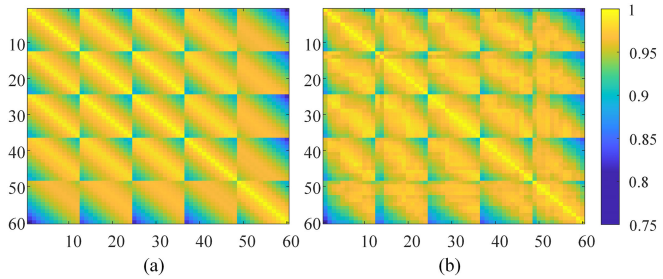


Fig. 9. Correlation matrix heatmaps of PV power in the summer case: (a) historical data, (b) forecasting scenarios.

VI. Conclusion

This paper develops the probabilistic forecasting model of PV power generation based on the MBLS and Copula theory. The quantile crossing problem encountered in probabilistic forecasting is thoroughly solved based on the proposed QRMBLS model. Moreover, the conditional spatio-temporal dependence of PV power generation is modeled using clustering-based Copula. The effectiveness of the forecasting method is validated on the actual data of PV units. The case studies show that the proposed marginal probabilistic forecasting method provides reliable and sharp predictive quantiles without crossing in a computationally efficient manner. Furthermore, the proposed spatio-temporal forecasting method accurately models the conditional dependence of multiple PV units and time steps, and it produces high-quality spatio-temporal PV power generation scenarios.

Future work could integrate numerical weather prediction data into the proposed method to predict PV power of longer time horizons. Moreover, advanced Copula functions such as vine Copulas would be applied to modeling the complicated dependence structure of renewable energy power generation.

Appendix

A. Proof of Lemma 1

By writing (10) in an element-wise manner, the expression of $\hat{Y}_n$ is obtained as

$$\begin{aligned}\hat{Y}_n &= r\left(\sum_{i=1}^{N_{\mathrm{g}}\times N_{\mathrm{f}}} F_{n,i}W_{\mathrm{N}i} + \sum_{j=1}^{N_{\mathrm{e}}} E_{n,j}W_{\mathrm{M}j}^2 + \beta\right)\\ &= r\left(\sum_{i=1}^{N_{\mathrm{g}}\times N_{\mathrm{f}}} F_{n,i}W_{\mathrm{N}i} + \sum_{j=1}^{N_{\mathrm{e}}} \zeta\left(\sum_{i=1}^{N_{\mathrm{g}}\times N_{\mathrm{f}}} F_{n,i}W_{\mathrm{E}i,j}\right.\right.\\ &\quad \left.\left. + X_{\mathrm{M}n}V_{\mathrm{M1},j}^2 + \beta_{\mathrm{E}j}\right)W_{\mathrm{M}j}^2 + \beta\right)\end{aligned} \tag{31}$$

Since $\boldsymbol{X}_{\mathrm{N}_{n_1}} = \boldsymbol{X}_{\mathrm{N}n_2}$, $F_{n,i}$ is viewed as constant according to (8). The partial derivative of $\hat{Y}_n$ with respect to $X_{\mathrm{M}n}$ is given

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

by:

$$\frac{\partial \hat{Y}_n}{\partial X_{\mathrm{M}n}} = r' \sum_{j=1}^{N_\mathrm{e}} \zeta' \boldsymbol{V}_{\mathrm{M1},j}^2 \boldsymbol{W}_{\mathrm{M}j}^2 \tag{32}$$

Since both $\zeta(\cdot)$ and $r(\cdot)$ are nondecreasing, their derivatives ζ' and r' are nonnegative. Hence, $\partial \hat{Y}_n / \partial X_{\mathrm{M}n} \geq 0$, which implies that $X_{\mathrm{M}n_1} \geq X_{\mathrm{M}n_2}$ leads to $\hat{Y}_{n_1} \geq \hat{Y}_{n_2}$. ■

B. Huber Approximations of Pinball Loss and Ramp Function

The approximated pinball loss function (4) is given by:

$$\rho_\tau^{(\mathrm{A})}(\varepsilon) = \begin{cases} \tau\varphi(\varepsilon), & \varepsilon \geq 0 \\ (1-\tau)\varphi(\varepsilon), & \varepsilon < 0 \end{cases} \tag{33}$$

and the approximated ramp function (11) is given by:

$$r^{(\mathrm{A})}(\pi) = \begin{cases} 0, & \pi < 0 \\ \varphi(\pi), & 0 \leq \pi < P_{\text{cap}} \\ P_{\text{cap}} - \varphi(P_{\text{cap}} + a - \pi), & P_{\text{cap}} \leq \pi \leq P_{\text{cap}} + a \\ P_{\text{cap}}, & \pi > P_{\text{cap}} + a \end{cases} \tag{34}$$

where the Huber function

$$\varphi(\varepsilon) = \begin{cases} \frac{\varepsilon^2}{2a}, & |\varepsilon| \leq a \\ |\varepsilon| - \frac{a}{2}, & |\varepsilon| > a \end{cases} \tag{35}$$

is a combination of the L1- and L2-norms. The Huber function guarantees differentiability by applying the L2-norm around the origin. As $a \to 0$, the approximation converges to the original function.

C. Gradient of Loss Function for MBLS

The gradient of loss function (15) is made up of derivates of $L^{(\mathrm{A})}$ with respect to different parameters of MBLS. For convenience, denote the first term in (15) as:

$$\hat{L}^{(\mathrm{A})} = \frac{1}{N} \sum_{n=1}^{N} \rho_{\tau_n}^{(\mathrm{A})}(Y_n - \hat{Y}_n^{(\mathrm{A})}) \tag{36}$$

Denote that $D_{\rho_\tau}(\varepsilon) = \partial \rho_\tau^{(\mathrm{A})} / \partial \varepsilon$, $D_r(\pi) = \partial r^{(\mathrm{A})} / \partial \pi$, $\varepsilon_n = Y_n - \hat{Y}_n$, and $\hat{Y}_n^{(\mathrm{A})} = r^{(\mathrm{A})}(\pi_n)$. Assume that θ is a parameter of MBLS, the derivative of $\hat{L}^{(\mathrm{A})}$ with respect to θ is derived according to the chain rule:

$$\begin{aligned} \frac{\partial \hat{L}^{(\mathrm{A})}}{\partial \theta} &= \frac{1}{N} \sum_{n=1}^{N} \frac{\partial}{\partial \theta} \rho_{\tau_n}^{(\mathrm{A})}(Y_n - \hat{Y}_n^{(\mathrm{A})}) \\ &= \frac{1}{N} \sum_{n=1}^{N} \frac{\partial \rho_{\tau_n}^{(\mathrm{A})}}{\partial \varepsilon_n} \frac{\partial \varepsilon_n}{\partial \hat{Y}_n^{(\mathrm{A})}} \frac{\partial \hat{Y}_n^{(\mathrm{A})}}{\partial \pi_n} \frac{\partial \pi_n}{\partial \theta} \\ &= \frac{1}{N} \sum_{n=1}^{N} -D_{\rho_{\tau_n}}(\varepsilon_n) D_r(\pi_n) \frac{\partial \pi_n}{\partial \theta} \end{aligned} \tag{37}$$

Note that the derivative of hyperbolic tangent function $\zeta(\cdot)$ satisfies that:

$$\frac{d\zeta(x)}{dx} = 1 - \zeta(x)^2 \tag{38}$$

Hence, $\partial \pi_n / \partial \theta$ is derived using backpropagation according to (31):

$$\begin{aligned} \frac{\partial \pi_n}{\partial W_{\mathrm{N}i}} &= F_{n,i} \\ \frac{\partial \pi_n}{\partial W_{\mathrm{M}j}} &= 2E_{n,j} W_{\mathrm{M}j} \\ \frac{\partial \pi_n}{\partial \beta} &= 1 \\ \frac{\partial \pi_n}{\partial V_{\mathrm{M1},j}} &= (1 - E_{n,j}^2) \cdot 2\tau_n V_{\mathrm{M1},j} \cdot W_{\mathrm{M}j}^2 \\ \frac{\partial \pi_n}{\partial \beta_{\mathrm{Ej}}} &= (1 - E_{n,j}^2) W_{\mathrm{M}j}^2 \end{aligned} \tag{39}$$

Hence, the gradients of loss function with respect to different parameters are given by:

$$\begin{aligned} \delta W_{\mathrm{N}i} = \frac{\partial L^{(\mathrm{A})}}{\partial W_{\mathrm{N}i}} &= -\frac{1}{N} \sum_{n=1}^{N} D_{\rho_{\tau_n}}(\varepsilon_n) D_r(\pi_n) F_{n,i} \\ &\quad + \lambda_{W_\mathrm{I}} \frac{2}{N_\mathrm{g} N_\mathrm{f}} W_{\mathrm{N}i} \end{aligned} \tag{40}$$

$$\begin{aligned} \delta W_{\mathrm{M}i} = \frac{\partial L^{(\mathrm{A})}}{\partial W_{\mathrm{M}i}} &= -\frac{2}{N} \sum_{n=1}^{N} D_{\rho_{\tau_n}}(\varepsilon_n) D_r(\pi_n) \\ &\quad \times E_{n,j} W_{\mathrm{M}j} + \lambda_{W_\mathrm{M}} \frac{2}{N_\mathrm{e}} W_{\mathrm{M}j} \end{aligned} \tag{41}$$

$$\delta \beta = \frac{\partial L^{(\mathrm{A})}}{\partial \beta} = -\frac{1}{N} \sum_{n=1}^{N} D_{\rho_{\tau_n}}(\varepsilon_n) D_r(\pi_n) \tag{42}$$

$$\begin{aligned} \delta V_{\mathrm{M1},j} = \frac{\partial L^{(\mathrm{A})}}{\partial V_{\mathrm{M1},j}} &= -\frac{2}{N} \sum_{n=1}^{N} \big[D_{\rho_{\tau_n}}(\varepsilon_n) D_r(\pi_n) \\ &\quad \cdot (1 - E_{n,j}^2) \tau_n V_{\mathrm{M1},j} W_{\mathrm{M}j}^2 \big] + \lambda_{V_\mathrm{M}} \frac{2}{N_\mathrm{e}} V_{\mathrm{M1},j} \end{aligned} \tag{43}$$

$$\begin{aligned} \delta \beta_{\mathrm{Ej}} = \frac{\partial L^{(\mathrm{A})}}{\partial \beta_{\mathrm{Ej}}} &= -\frac{1}{N} \sum_{n=1}^{N} D_{\rho_{\tau_n}}(\varepsilon_n) D_r(\pi_n) \\ &\quad \times (1 - E_{n,j}^2) W_{\mathrm{M}j}^2 \end{aligned} \tag{44}$$

REFERENCES

[1] C. Li *et al.*, "Optimal planning of islanded integrated energy system with solar-biogas energy supply," *IEEE Trans. Sustain. Energy*, vol. 11, no. 4, pp. 2437–2448, Oct. 2020.

[2] N. Haghdadi, A. Bruce, I. MacGill, and R. Passey, "Impact of distributed photovoltaic systems on zone substation peak demand," *IEEE Trans. Sustain. Energy*, vol. 9, no. 2, pp. 621–629, Apr. 2018.

[3] J. Antonanzas, N. Osorio, R. Escobar, R. Urraca, F. J. Martinez-de-Pison, and F. Antonanzas-Torres, "Review of photovoltaic power forecasting," *Sol. Energy*, vol. 136, pp. 78–111, Oct. 2016.

[4] F. Golestaneh, P. Pinson, and H. B. Gooi, "Very short-term nonparametric probabilistic forecasting of renewable energy generation— With application to solar energy," *IEEE Trans. Power Syst.*, vol. 31, no. 5, pp. 3850–3863, Sep. 2016.

[5] Z. Wang, W. Wang, C. Liu, Z. Wang, and Y. Hou, "Probabilistic forecast for multiple wind farms based on regular vine copulas," *IEEE Trans. Power Syst.*, vol. 33, no. 1, pp. 578–589, Jan. 2018.

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.

[6] M. Kakimoto, Y. Endoh, H. Shin, R. Ikeda, and H. Kusaka, "Probabilistic solar irradiance forecasting by conditioning joint probability method and its application to electric power trading," *IEEE Trans. Sustain. Energy*, vol. 10, no. 2, pp. 983–993, Apr. 2019.
[7] M. Sun, T. Zhang, Y. Wang, G. Strbac, and C. Kang, "Using Bayesian deep learning to capture uncertainty for residential net load forecasting," *IEEE Trans. Power Syst.*, vol. 35, no. 1, pp. 188–201, Jan. 2020.
[8] A. Bracale, P. Caramia, P. De Falco, and T. Hong, "Multivariate quantile regression for short-term probabilistic load forecasting," *IEEE Trans. Power Syst.*, vol. 35, no. 1, pp. 628–638, Jan. 2020.
[9] C. Wan, J. Lin, Y. Song, Z. Xu, and G. Yang, "Probabilistic forecasting of photovoltaic generation: An efficient statistical approach," *IEEE Trans. Power Syst.*, vol. 32, no. 3, pp. 2471–2472, May 2017.
[10] C. Wan, J. Lin, J. Wang, Y. Song, and Z. Y. Dong, "Direct quantile regression for nonparametric probabilistic forecasting of wind power generation," *IEEE Trans. Power Syst.*, vol. 32, no. 4, pp. 2767–2778, Jul. 2017.
[11] Y. Wang, N. Zhang, Y. Tan, T. Hong, D. S. Kirschen, and C. Kang, "Combining probabilistic load forecasts," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 3664–3674, Jul. 2019.
[12] W. Zhang, H. Quan, and D. Srinivasan, "An improved quantile regression neural network for probabilistic load forecasting," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 4425–4434, Jul. 2019.
[13] W. Zhang, H. Quan, O. Gandhi, R. Rajagopal, C. Tan, and D. Srinivasan, "Improving probabilistic load forecasting using quantile regression NN with skip connections," *IEEE Trans. Smart Grid*, vol. 11, no. 6, pp. 5442–5450, Nov. 2020.
[14] Y. Wang, D. Gan, M. Sun, N. Zhang, Z. Lu, and C. Kang, "Probabilistic individual load forecasting using pinball loss guided LSTM," *Appl. Energy*, vol. 235, pp. 10–20, Feb. 2019.
[15] V. Chernozhukov, I. Fernández-Val, and A. Galichon, "Quantile and probability curves without crossing," *Econometrica*, vol. 78, no. 3, pp. 1093–1125, May 2010.
[16] K. Qu, G. Si, Z. Yang, Y. Huang, and P. Li, "Correlation modeling of multiple wind farms based on piecewise cloud representation and regular vine copulas," *Energy Rep.*, vol. 6, no. 9, pp. 289–297, Dec. 2020.
[17] J. Dowell and P. Pinson, "Very-short-term probabilistic wind power forecasts by sparse vector autoregression," *IEEE Trans. Smart Grid*, vol. 7, no. 2, pp. 763–770, Mar. 2016.
[18] C. Lucheroni, J. Boland, and C. Ragno, "Scenario generation and probabilistic forecasting analysis of spatio-temporal wind speed series with multivariate autoregressive volatility models," *Appl. Energy*, vol. 239, pp. 1226–1241, Apr. 2019.
[19] R. Becker, "Generation of time-coupled wind power infeed scenarios using pair-copula construction," *IEEE Trans. Sustain. Energy*, vol. 9, no. 3, pp. 1298–1306, Jul. 2018.
[20] J. Toubeau, J. Bottieau, F. Vallée, and Z. De Grève, "Deep learning-based multivariate probabilistic forecasting for short-term scheduling in power markets," *IEEE Trans. Power Syst.*, vol. 34, no. 2, pp. 1203–1215, Mar. 2019.
[21] M. Sun, C. Feng, and J. Zhang, "Conditional aggregated probabilistic wind power forecasting based on spatio-temporal correlation," *Appl. Energy*, vol. 256, Dec. 2019, Art. no. 113842.
[22] C. L. P. Chen and Z. Liu, "Broad learning system: An effective and efficient incremental learning system without the need for deep architecture," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 1, pp. 10–24, Jan. 2018.
[23] C. L. P. Chen, Z. Liu, and S. Feng, "Universal approximation capability of broad learning system and its structural variations," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 4, pp. 1191–1204, Apr. 2019.
[24] A. J. Cannon, "Non-crossing nonlinear regression quantiles by monotone composite quantile regression neural network, with application to rainfall extremes," *Stochastic Environ. Res. Risk Assessment*, vol. 32, pp. 3207–3225, Jun. 2018.
[25] R. Koenker and G. Bassett, "Regression quantiles," *Econometrica*, vol. 46, no. 1, pp. 33–50, Jan. 1978.
[26] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Representations*, 2015, pp. 1–15.
[27] T. Kohonen, "The self-organizing map," *Proc. IEEE*, vol. 78, no. 9, pp. 1464–1480, Sep. 1990.
[28] B. Choroś, R. Ibragimov, and E. Permiakova, "Copula estimation," in *Copula Theory and Its Applications*. Berlin, Germany: Springer, 2010, pp. 77–91.
[29] G. Papaefthymiou and D. Kurowicka, "Using copulas for modeling stochastic dependence in power system uncertainty analysis," *IEEE Trans. Power Syst.*, vol. 24, no. 1, pp. 40–49, Feb. 2009.
[30] F. Rodrigues and F. C. Pereira, "Beyond expectation: Deep joint mean and quantile regression for spatiotemporal problems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 12, pp. 5377–5389, Dec. 2020.
[31] DKA Solar Centre. Data Download. Aug. 1, 2021. [Online]. Available: http://dkasolarcentre.com.au/download?location=yulara
[32] P. Singla, D. Manoj, and S. Saroha, "A comprehensive review and analysis of solar forecasting techniques," *Front. Energy*, vol. 15, pp. 1–37, Mar. 2021.
[33] University of Oregon Solar Radiation Monitoring Laboratory. Download Solar data. Jan. 15, 2022. [Online]. Available: http://solardata.uoregon.edu/SolarData.html
[34] N. Zhou, "Spatio-Temporal forecasting based on MBLS," [Online]. Available: https://github.com/ZhouNan2015/SpatioTemporalForecastingBasedOnMBLS
[35] A. J. Cannon, "Quantile regression neural networks: Implementation in r and application to precipitation downscaling," *Comput. Geosci.*, vol. 37, no. 9, pp. 1277–1284, Sep. 2011.



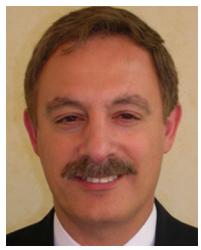
Nan Zhou received the B.S. degree in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2019. He is currently working towards the M.S. degree at the Department of Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China. His research interests include renewable energy power generation forecasting and application of machine learning in power systems.



Xiaoyuan Xu (Member, IEEE) received the B.S. and Ph.D. degrees in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2010 and 2016, respectively. He was a visiting scholar with the Robert W. Galvin Center for Electricity Innovation, Illinois Institute of Technology, Chicago, IL, USA, from 2017 to 2018. He is currently an Associate Professor of Electrical Engineering with Shanghai Jiao Tong University. His research interests include power system uncertainty analysis and power system optimization.



Zheng Yan (Member, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1984 and the M.S. and Ph.D. degrees from Tsinghua University, Beijing, China, in 1987 and 1991, respectively, all in electrical engineering. He is currently a Professor of Electrical Engineering with Shanghai Jiao Tong University. His current research interests include application of optimization theory to power systems, power markets, and dynamic security assessment.



Mohammad Shahidehpour (Life Fellow, IEEE) received the Honorary Doctorate degree in electrical engineering from the Polytechnic University of Bucharest, Bucharest, Romania. He is the Bodine Chair Professor and the Director with the Robert W. Galvin Center for Electricity Innovation, Illinois Institute of Technology, Chicago, IL, USA. He is a member of the US National Academy of Engineering and a Fellow of the American Association for the Advancement of Science.

Authorized licensed use limited to: ITA - Instituto Tecnológico de Aeronáutica. Downloaded on May 26,2026 at 00:55:39 UTC from IEEE Xplore. Restrictions apply.